








Time Series Analysis

Chapter 5: Conditional volatility: ARCH and GARCH



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Acronyms used in this material (1/2)

- **ACF**: Autocorrelation Function
- **AI**: Artificial Intelligence
- **AIC**: Akaike Information Criterion
- **AR**: AutoRegressive (model)
- **AR-GARCH**: AR model for the mean with GARCH errors
- **ARCH**: AutoRegressive Conditional Heteroskedasticity
- **ARCH-LM**: ARCH Lagrange Multiplier (test)
- **ARMA**: AutoRegressive Moving Average (model)
- **ARMA-GARCH**: ARMA model for the mean with GARCH errors
- **BET**: Bucharest Exchange Trading (index)
- **BIC**: Bayesian Information Criterion
- **BNR**: Banca Națională a României (the National Bank of Romania)
- **BVB**: Bursa de Valori București (the Bucharest Stock Exchange)
- **COVID-19**: COronaVIrus Disease 2019
- **DM**: Diebold–Mariano (test)
- **EGARCH**: Exponential GARCH
- **EWMA**: Exponentially Weighted Moving Average

Acronyms used in this material (2/2)

- **GARCH**: Generalised AutoRegressive Conditional Heteroskedasticity
- **GARCH-N**: GARCH with Normal innovations
- **GJR**: Glosten–Jagannathan–Runkle (GARCH model)
- **GJR-GARCH**: Glosten–Jagannathan–Runkle GARCH
- **HAC**: Heteroskedasticity and Autocorrelation Consistent (standard errors)
- **IGARCH**: Integrated GARCH
- **LM**: Lagrange Multiplier
- **LR**: Likelihood Ratio (test)
- **MA**: Moving Average
- **MLE**: Maximum Likelihood Estimation
- **NIC**: News Impact Curve
- **OLS**: Ordinary Least Squares
- **PACF**: Partial AutoCorrelation Function
- **QLIKE**: Quasi-LIKElihood (loss function)
- **QMLE**: Quasi-Maximum Likelihood Estimation
- **QQ**: Quantile–Quantile (plot)
- **SE**: Standard Error
- **US**: United States
- **VAR**: Vector AutoRegression

Today's question and route

- **Question**: if tomorrow's return cannot be predicted, can tomorrow's **risk** be predicted?
  - Chapters 1–3 modelled the conditional **mean** of a series; this chapter models its conditional **variance**
- **Route** of the chapter
  - stylised facts of returns; the ACF of squared returns and the ARCH-LM test
  - conditional mean and conditional variance; ARCH(q) and GARCH(1,1); persistence, IGARCH and EWMA
  - estimation by maximum likelihood with the `arch` package; Student-t innovations; ARMA-GARCH models
  - asymmetry (GJR, EGARCH), diagnostics, variance forecasts and their evaluation, VaR 1%
- Data: S&P 500, BET, EUR/RON (BNR reference rate) and Bitcoin, daily; Seminar 5 comes before this lecture

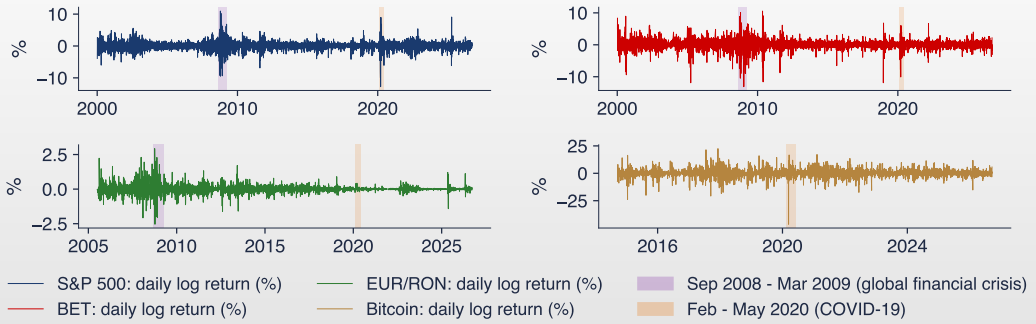
Learning outcomes

- List the stylised facts of financial returns and test for ARCH effects with the ACF of squared returns and the ARCH-LM test
- Distinguish the conditional from the unconditional variance, and explain why returns can be uncorrelated yet dependent
- Write ARCH(q), GARCH(1,1) and ARMA-GARCH models; compute the persistence, the long-run variance and the half-life
- Estimate GARCH models by maximum likelihood in Python, choose the innovation distribution and check the standardised residuals
- Measure asymmetry with GJR-GARCH, EGARCH and the news impact curve
- Forecast the variance over several horizons, evaluate the forecasts with QLIKE and compute a one-day VaR 1%

Reading and tools

- Textbook: Huang and Petukhina (2022), Ch. 6 (financial time series: ARCH and GARCH models)
  - further reading: Tsay (2010), Ch. 3; Franke, Härdle and Hafner (2019), Ch. 13
- Theory: Hamilton (1994), Ch. 21 (time series models of heteroskedasticity)
- Python Quantlets of this chapter: Quantlets/Ch_05
  - estimation with the package `arch` (install it in Colab with `pip install arch`); tests with `statsmodels`
- Lecture notebook: open in Google Colab
- Video course: Applied Time Series Analysis with Python

Four markets, one pattern



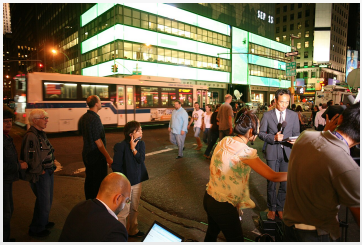
- Daily log returns $r_t = 100(\ln P_t - \ln P_{t-1})$, in %, to 18 September 2026: S&P 500 and BET since 2000, EUR/RON (BNR reference rate) since July 2005, Bitcoin since September 2014
- Shaded: the global financial crisis (September 2008 – March 2009) and the COVID-19 crash (February – May 2020)

TSA_ch5_stylised_facts

Interpreting the four series

- Calm years alternate with storms: large changes follow large changes, of either sign
  - this is **volatility clustering**, first described by Mandelbrot (1963)
- S&P 500: daily standard deviation 0.42% in 2017, 3.52% from September 2008 to March 2009 ($\times$8.3) and 3.70% in February–May 2020 ($\times$8.8)
- EUR/RON: 0.78% in the 2008 crisis, only 0.10% in 2020: the BNR kept the leu almost fixed
- A model with a constant variance (white noise, Chapter 1) describes none of these series

Two storms: 2008 and 2020



Lehman Brothers headquarters, New York, 15 September 2008

Photo: Robert Scoble (2008); CC BY 2.0; Wikimedia Commons



The Financial District of New York, 25 March 2020

Photo: Billie Grace Ward (2020); CC BY 2.0; Wikimedia Commons

- 15 September 2008: Lehman Brothers files for bankruptcy; March 2020: the COVID-19 pandemic closes economies
- In both episodes calm returned only after months: a shock to volatility is persistent

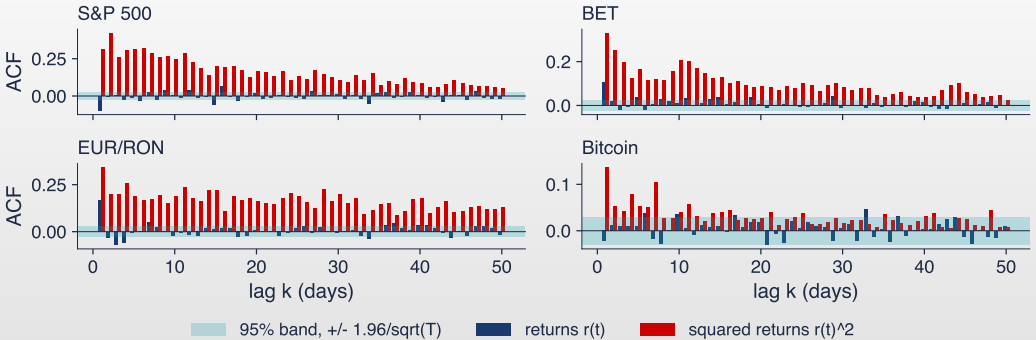
Stylised facts in numbers

	T	s.d.	skew.	kurt.	$\hat{\rho}_1(r)$	$\hat{\rho}_1(r^2)$	$Q(10)$, r	$Q(10)$, r^2	LM(5)
S&P 500	6 714	1.21	−0.35	13.7	−0.10	0.31	96	6133	1684
BET	6 670	1.40	−0.65	14.1	0.11	0.33	108	2440	982
EUR/RON	5 352	0.28	0.73	18.1	0.17	0.34	222	2420	871
Bitcoin	4 384	3.49	−0.70	15.0	−0.02	0.14	20	213	115

- **Heavy tails**: kurtosis between 13.7 and 18.1, far above 3, the value of the Normal distribution
  - kurtosis $K = E[(r-\mu)^4]/\sigma^4$; skewness $E[(r-\mu)^3]/\sigma^3$ (Chapter 1)
- **Little autocorrelation in r_t**, strong autocorrelation in r_t^2: $Q(10)$ of the squares is 11–64 times larger
- $Q(10)$: the statistic of Ljung and Box (1978) (Chapter 1), $\chi^2(10)$, 5% value 18.31; LM(5): the ARCH-LM test (two slides ahead), $\chi^2(5)$, 5% value 11.07

TSA_ch5_stylised_facts

The sign is unpredictable, the size is not



- Sample ACF (autocorrelation function, Chapter 1) of r_t and of r_t^2, lags 1–50, with the band $\pm 1.96/\sqrt{T}$ of an i.i.d. (independent and identically distributed) series

TSA_ch5_stylised_facts

Interpreting the two ACFs

- Returns: almost all autocorrelations inside the band (BET: $\hat{\rho}_1 = 0.11$, a thin market; Chapter 2)
  - the direction of tomorrow's move is almost unpredictable
- Squared returns: S&P 500 0.31 at lag 1 and still 0.05 at lag 50; 50 of 50 lags outside the band
  - the size of tomorrow's move is predictable: a large move today makes a large move tomorrow more likely
- Uncorrelated is not the same as independent: r_t is close to white noise but not i.i.d. (Chapter 1)
- Bitcoin: weaker but significant ($\hat{\rho}_1(r_t^2) = 0.14$): a few huge days dominate the squares

Testing for ARCH effects

- **ARCH effects**: the conditional variance depends on the past (the name comes from the ARCH model, Section 3)
- **Ljung–Box on squares** (the test of McLeod and Li (1983)): $Q(m) = T(T+2)\sum_{k=1}^{m} \hat{\rho}_k^2(\hat{\varepsilon}^2)/(T-k)$
  - $\hat{\rho}_k(\hat{\varepsilon}^2)$: the sample autocorrelation of the squared residuals at lag k; m: the number of lags; T: the number of observations
  - $\hat{\varepsilon}_t$: residuals of the mean model (for returns, $r_t - \bar{r}$); under H_0 (no ARCH effects), $Q(m) \sim \chi^2(m)$
- **ARCH-LM test** Engle (1982): LM (Lagrange multiplier) test from the auxiliary regression
  - $\hat{\varepsilon}_t^2 = b_0 + b_1\hat{\varepsilon}_{t-1}^2 + \cdots + b_q\hat{\varepsilon}_{t-q}^2 + u_t$, estimated by OLS (ordinary least squares)
  - $b_0, \ldots, b_q$: the coefficients; q: the number of lagged squares; R^2: the coefficient of determination of this regression
  - H_0: $b_1 = \cdots = b_q = 0$; LM $= nR^2 \sim \chi^2(q)$, $n =$ number of observations in the regression
- Both tests are run on the residuals of the mean model, before and after fitting a volatility model

Worked example: ARCH-LM on the BET

- Mean model: AR(1) for the BET returns (Chapter 2), $\hat{\phi} = 0.109$; residuals $\hat{\varepsilon}_t$; $q = 5$
- Auxiliary regression (OLS)
  - $\hat{\varepsilon}_t^2 = 0.862 + 0.276\,\hat{\varepsilon}_{t-1}^2 + 0.099\,\hat{\varepsilon}_{t-2}^2 + 0.088\,\hat{\varepsilon}_{t-3}^2 + (-0.015)\,\hat{\varepsilon}_{t-4}^2 + 0.102\,\hat{\varepsilon}_{t-5}^2$
  - $R^2 = 0.1526$, $n = 6\,664$
- Step by step: $\text{LM} = nR^2 = 6\,664 \times 0.1526 = 1017$; 5% critical value $\chi^2_{0.95}(5) = 11.07$
  - $1017 \gg 11.07$: reject H_0; the variance of the BET depends on the last five squared shocks
- Interpretation: $R^2 = 0.1526$ is small, but for squared returns it is a strong signal; the model of the variance is the next step

TSA_ch5_mean_model

Recap: Stylised facts

- Heavy tails, volatility clustering, little autocorrelation in r_t, strong autocorrelation in r_t^2
- ARCH effects: Ljung–Box on squared residuals, ARCH-LM $= nR^2 \sim \chi^2(q)$
- All four series reject H_0 by a wide margin

Conditioning on the past

- $\mathcal{F}_{t-1}$: the information available at the end of day $t-1$ (all past values)
  - conditional expectation $E[X \mid \mathcal{F}_{t-1}]$: the best forecast of X given the past (Chapter 2)
- **Conditional mean**: $\mu_t = E[r_t \mid \mathcal{F}_{t-1}]$
  - an ARMA model (Chapter 2) is a model for μ_t; for daily returns μ_t is small and almost constant
- **Conditional variance**: $\sigma_t^2 = \mathrm{Var}(r_t \mid \mathcal{F}_{t-1}) = E[(r_t - \mu_t)^2 \mid \mathcal{F}_{t-1}]$
  - σ_t is the **conditional volatility**: known at $t-1$, it changes from day to day
  - **conditional heteroskedasticity**: a conditional variance that is not constant
- **Unconditional variance** $\mathrm{Var}(r_t)$: one number for the whole sample; it can be constant while σ_t^2 moves

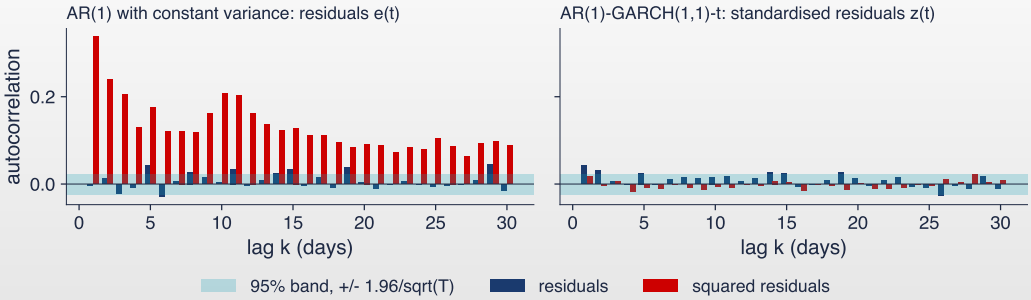
The basic decomposition

- Every model of this chapter: $r_t = \mu_t + \varepsilon_t, \quad \varepsilon_t = \sigma_t z_t$
  - z_t: the **innovations**, i.i.d. with mean 0 and variance 1 (Normal, Student-t, ...)
  - ε_t: the **shock**; μ_t and σ_t depend only on the past
- Consequences
  - $E[\varepsilon_t \mid \mathcal{F}_{t-1}] = \sigma_t E[z_t] = 0$: the shocks are a **martingale difference**, hence uncorrelated (white noise)
  - $\mathrm{Var}(\varepsilon_t \mid \mathcal{F}_{t-1}) = \sigma_t^2$: the variance is predictable
  - the ε_t are uncorrelated but not independent: ε_t^2 is correlated with ε_{t-1}^2
- Two models, one for each moment: ARMA for μ_t (Chapter 2), GARCH for σ_t^2 (this chapter); together: **ARMA-GARCH**

Worked example: conditional and unconditional variance

- A market has calm days ($\sigma_t^2 = 0.5$) and stormy days ($\sigma_t^2 = 4.5$), each with probability $1/2$; the mean is 0
  - law of total variance: $\mathrm{Var}(r_t) = E[\mathrm{Var}(r_t \mid \mathcal{F}_{t-1})] + \mathrm{Var}(E[r_t \mid \mathcal{F}_{t-1}])$
  - step by step: $0.5 \times 0.5 + 0.5 \times 4.5 + 0 = 2.50$; unconditional standard deviation 1.58%
- The unconditional variance is an average; the conditional variance tells which day we are in
  - a calm day: $\sigma_t = 0.71\%$; a stormy day: $\sigma_t = 2.12\%$; the average 1.58% fits neither
- With Normal z_t, the mixture has kurtosis $3E[\sigma_t^4]/(E[\sigma_t^2])^2 = 4.92$: a time-varying variance creates heavy tails

The mean model leaves ARCH effects



- BET: left, an AR(1) for the mean (order chosen by BIC among AR(0)–AR(5): $p = 1$), constant variance; right, the same AR(1) with a GARCH(1,1)-t variance (Section 6)

TSA_ch5_mean_model

Interpreting the residual ACFs

- Left: the residuals look almost like white noise, the Box–Jenkins check of Chapter 2 would be nearly passed
  - but their squares: $\hat{\rho}_1 = 0.34$, $Q(10) = 2496$: strong ARCH effects
- Right: after dividing by $\hat{\sigma}_t$, the squares are clean: $\hat{\rho}_1 = 0.018$, $Q(10) = 7.2$ (p $= 0.705$), ARCH-LM 4.8 (p $=$ 0.446)
  - some autocorrelation in the level remains ($Q(10) = 28.6$): the BET mean has more structure than an AR(1)
- Lesson: check both $\hat{\varepsilon}_t$ and $\hat{\varepsilon}_t^2$; a white-noise check on $\hat{\varepsilon}_t$ alone misses the variance

Recap: Conditional mean and variance

- $r_t = \mu_t + \sigma_t z_t$: μ_t from an ARMA model, σ_t^2 from a volatility model
- Shocks are uncorrelated but dependent through their squares
- Unconditional variance = average of the conditional variances; the mixture has heavy tails

1982: Robert Engle and ARCH

- Engle (1982): **ARCH** (autoregressive conditional heteroskedasticity): today's variance depends on yesterday's squared shocks
  - first application: the variance of inflation in the United Kingdom, not a financial market
- Nobel Prize 2003: "for methods of analyzing economic time series with time-varying volatility (ARCH)"
  - shared with Clive Granger (Chapter 3: spurious regression; Chapter 7: cointegration); Nobel lecture: Engle (2004)



Robert F. Engle (2022)

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Stockholm Concert Hall, venue of the Nobel Prize ceremony

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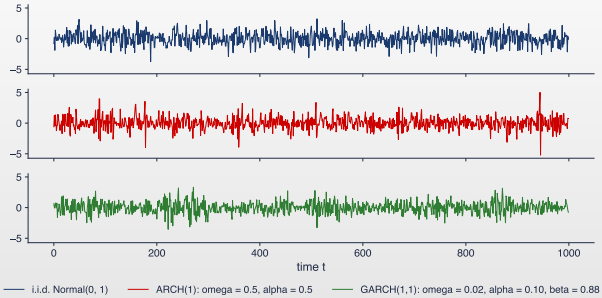
ARCH(1): definition

- $\varepsilon_t = \sigma_t z_t, \quad \sigma_t^2 = \omega + \alpha \varepsilon_{t-1}^2, \quad \omega > 0,\ \alpha \geq 0$
  - ω: the minimum level of the variance; α: how strongly the variance reacts to yesterday's squared shock
  - $\omega > 0$ and $\alpha \geq 0$ keep the variance positive
  - a large shock yesterday, of either sign, raises today's variance
- "Autoregressive": ε_t^2 follows an AR(1) model (Chapter 2)
  - write $\varepsilon_t^2 = \sigma_t^2 + v_t$, with $v_t = \sigma_t^2(z_t^2 - 1)$ and $E[v_t \mid \mathcal{F}_{t-1}] = 0$
  - then $\varepsilon_t^2 = \omega + \alpha \varepsilon_{t-1}^2 + v_t$: an AR(1) for the squared shocks, with white-noise errors v_t
- Hence the ACF of ε_t^2 is $\rho_k = \alpha^k$ (when the fourth moment exists): clustering, but a fast decay

ARCH(1): properties

- $E[\varepsilon_t] = 0$ and $\text{Cov}(\varepsilon_t, \varepsilon_{t-k}) = 0$ for $k \neq 0$: white noise
- **Unconditional variance** (stationary if $\alpha < 1$): take expectations in $\sigma_t^2 = \omega + \alpha\varepsilon_{t-1}^2$
  - $\bar{\sigma}^2 = E[\varepsilon_t^2] = \omega + \alpha\bar{\sigma}^2$, so $\bar{\sigma}^2 = \omega/(1-\alpha)$
- **Kurtosis** with Normal z_t (if $3\alpha^2 < 1$): $K = 3\,\dfrac{1-\alpha^2}{1-3\alpha^2} > 3$
  - Normal innovations, yet heavy-tailed shocks; if $\alpha \geq 1/\sqrt{3} = 0.577$, the fourth moment is infinite
- **Worked example**: $\omega = 0.5$, $\alpha = 0.5$
  - $\bar{\sigma}^2 = 0.5/0.5 = 1.00$; $K = 3 \times 0.75/0.25 = 9.0$
  - after a shock $\varepsilon_{t-1} = 2$: $\sigma_t^2 = 0.5 + 0.5 \times 4 = 2.50$, $\sigma_t = 1.58$: the variance more than doubles

Simulated paths: i.i.d., ARCH(1) and GARCH(1,1)



- Three series of 1000 values with unconditional variance 1: i.i.d. Normal; ARCH(1) with $\omega = \alpha = 0.5$; GARCH(1,1) with $\omega = 0.02$, $\alpha = 0.10$, $\beta = 0.88$ (next section)
- Sample kurtosis: i.i.d. 3.12, ARCH 4.91, GARCH 3.74 (theory: 3, 9.0, 6.06): the fourth moment converges slowly
- Interpretation: ARCH(1) gives isolated spikes; GARCH(1,1) gives long calm and long agitated periods, as in real returns

TSA_ch5_simulation_likelihood

ARCH(q) and its limits

- **ARCH(q)**: $\sigma_t^2 = \omega + \alpha_1 \varepsilon_{t-1}^2 + \cdots + \alpha_q \varepsilon_{t-q}^2$, $\alpha_i \geq 0$; stationary if $\sum_i \alpha_i < 1$
  - a shock affects the variance for exactly q days; real clustering lasts for months

S&P 500, Normal z_t	parameters	log-likelihood	AIC	BIC	$\sum \alpha_i$ $(+\beta)$
ARCH(1)	3	−10305.5	20617.0	20637.5	0.394
ARCH(5)	7	−9340.5	18695.1	18742.7	0.816
ARCH(10)	12	−9227.6	18479.3	18561.0	0.849
GARCH(1,1)	4	−9218.0	**18444.0**	**18471.2**	0.980

- AIC (Akaike) and BIC (Bayesian information criterion): AIC $= -2\ell + 2k$, BIC $= -2\ell + k \ln T$ (Chapter 2), ℓ: the maximised log-likelihood, k: parameters; smaller is better
- Interpretation: ARCH needs ten lags and still loses to GARCH(1,1), which has four parameters: the motivation for GARCH

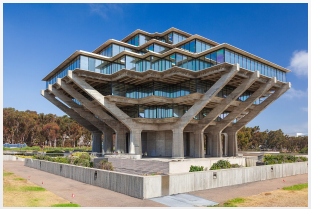
TSA_ch5_garch_estimation

Recap: The ARCH model

- $\sigma_t^2 = \omega + \sum_i \alpha_i \varepsilon_{t-i}^2$: an AR model for the squared shocks
- Unconditional variance $\omega/(1 - \sum \alpha_i)$; kurtosis above 3 even with Normal innovations
- Real clustering needs many lags: GARCH replaces them with one extra parameter

1986: Tim Bollerslev and GARCH

- Bollerslev (1986), a doctoral student of Engle at the University of California, San Diego: add yesterday's variance to the equation
  - **GARCH** (generalised ARCH): a few parameters replace a long ARCH(q)
- Student-t innovations for returns: Bollerslev (1987)
- Still the benchmark: Hansen and Lunde (2005) compared 330 volatility models and found GARCH(1,1) hard to beat on exchange rates



Geisel Library, University of California, San Diego

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GARCH(1,1): definition

- $\varepsilon_t = \sigma_t z_t, \quad \sigma_t^2 = \omega + \alpha\,\varepsilon_{t-1}^2 + \beta\,\sigma_{t-1}^2, \quad \omega > 0,\ \alpha \geq 0,\ \beta \geq 0$
  - GARCH(p,q) adds more lags of both kinds; in practice (1,1) is almost always enough
- Reading the parameters
  - α: the **reaction** to yesterday's news; β: the **memory** of the variance
  - $\alpha + \beta$: the **persistence**; ω sets the long-run level
- By recursion: an ARCH(∞) with geometric weights
  - $\sigma_t^2 = \dfrac{\omega}{1-\beta} + \alpha \sum_{j=0}^{\infty} \beta^j\, \varepsilon_{t-1-j}^2$: all past shocks matter, with weights that decay like β^j

GARCH(1,1) as an ARMA(1,1) for the squared shocks

- As for ARCH, write $\varepsilon_t^2 = \sigma_t^2 + v_t$ and substitute $\sigma_t^2 = \varepsilon_t^2 - v_t$ in the equation
  - $\varepsilon_t^2 = \omega + (\alpha + \beta)\,\varepsilon_{t-1}^2 + v_t - \beta\, v_{t-1}$
  - an ARMA(1,1) (Chapter 2) with AR coefficient $\alpha + \beta$ and MA coefficient $-\beta$
- Consequences
  - **covariance stationary** if $\alpha + \beta < 1$ (the AR root outside the unit circle)
  - the ACF of ε_t^2 decays like $(\alpha + \beta)^k$: slow when the persistence is close to 1, as in the ACF of real squared returns
  - identification by the ACF and the PACF (partial autocorrelation function) of ε_t^2 is possible in principle, but in practice we use maximum likelihood (Section 5)

Long-run variance, kurtosis, persistence and half-life

- **Long-run (unconditional) variance**: $\bar{\sigma}^2 = \omega/(1-\alpha-\beta)$; annualised volatility $\sqrt{252\,\bar{\sigma}^2}$ for 252 trading days
  - kurtosis with Normal z_t: $K = 3\,\dfrac{1-(\alpha+\beta)^2}{1-(\alpha+\beta)^2-2\alpha^2} > 3$; $\alpha = 0.10$, $\beta = 0.88$: $K = 6.06$
- A deviation from $\bar{\sigma}^2$ shrinks by the factor $\alpha+\beta$ each day (Section 9): $E_t[\sigma^2_{t+h}] - \bar{\sigma}^2 = (\alpha+\beta)^{h-1}(\sigma^2_{t+1} - \bar{\sigma}^2)$
  - $E_t[\cdot]$: the expectation given the information of day t; σ^2_{t+1}: tomorrow's variance, already known at t
  - **half-life**: the number of days after which half of a variance shock is gone, $h_{1/2} = \ln 0.5/\ln(\alpha+\beta)$
  - $\alpha+\beta = 0.90$: 6.6 days; 0.98: 34.3 days; 0.99: 69.0 days: small changes near 1 matter a lot

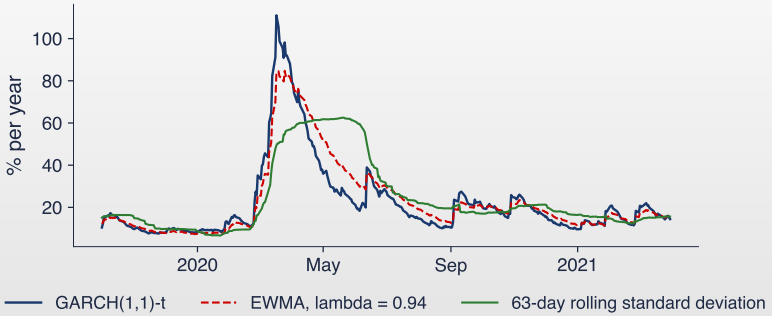
Worked example: one GARCH(1,1) step

- Daily returns in %: $\omega = 0.02$, $\alpha = 0.10$, $\beta = 0.88$, $\mu = 0$
- Long-run quantities
  - persistence $\alpha + \beta = 0.98$; $\bar{\sigma}^2 = 0.02/(1 - 0.98) = 1.00$; annualised volatility $\sqrt{252 \times 1.00} = 15.9\%$
  - half-life $\ln 0.5 / \ln 0.98 = 34.3$ days
- Today: $\sigma_t^2 = 1.2$ and the return is $r_t = -3\%$
  - tomorrow: $\sigma_{t+1}^2 = 0.02 + 0.10 \times 9 + 0.88 \times 1.2 = 1.976$, $\sigma_{t+1} = 1.41\%$
  - the shock raised the variance from 1.2 to almost 2; it then decays towards 1.00 with the factor 0.98 per day

IGARCH and the link with EWMA

- **IGARCH** (integrated GARCH) Engle and Bollerslev (1986): $\alpha + \beta = 1$: a unit root in the ARMA(1,1) for ε_t^2 (Chapter 3)
  - shocks to the variance never die out: no half-life, no finite unconditional variance
- **EWMA** (exponentially weighted moving average) of J.P. Morgan/Reuters (1996): $\sigma_t^2 = \lambda\sigma_{t-1}^2 + (1-\lambda)r_{t-1}^2$
  - $\lambda \in (0,1)$: the weight of yesterday's variance; $1-\lambda$: the weight of yesterday's squared return
  - simple exponential smoothing (Chapter 0) applied to r_t^2, with smoothing constant $1-\lambda$
  - an IGARCH with $\omega = 0$, $\alpha = 1-\lambda$, $\beta = \lambda$; RiskMetrics uses $\lambda = 0.94$ for daily data (half-life of the weights 11.2 days)
- **Question for the room**: right after a crash, an analyst forecasts the volatility one year ahead with EWMA. What goes wrong?
  - **Answer**: EWMA keeps today's crisis level for the whole year; GARCH lets it decay towards the long-run level

GARCH, EWMA and a rolling window in 2020



- S&P 500, annualised volatility: GARCH(1,1)-t estimated on all data, EWMA with $\lambda = 0.94$, and the standard deviation of the last 63 days (Chapter 0: a moving average)

TSA_ch5_volatility_markets

Interpreting the three estimates

- Peaks: GARCH 111% (17 March 2020), EWMA 85% (25 March 2020), 63-day window 63% (20 May 2020)
- The rolling window reacts late and falls abruptly when the crash leaves the window (the "ghost" of the crash)
  - on 30 June 2020: GARCH 28%, EWMA 30%, window 32%
- GARCH reacts fastest (α larger than $1 - \lambda$) and decays fastest (mean reversion); EWMA lies in between

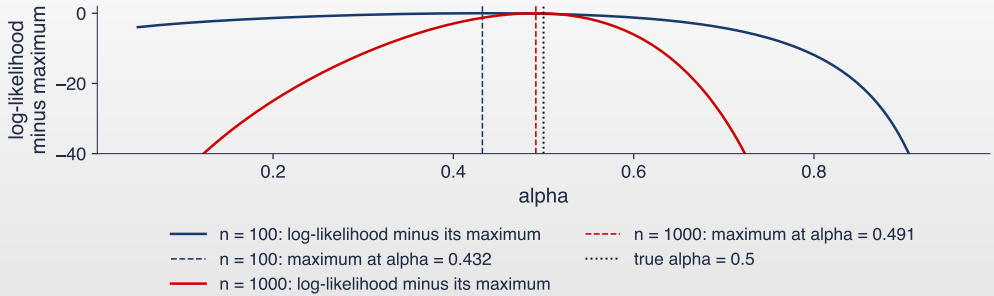
Recap: The GARCH(1,1) model

- $\sigma_t^2 = \omega + \alpha\varepsilon_{t-1}^2 + \beta\sigma_{t-1}^2$: reaction α, memory β; an ARMA(1,1) for ε_t^2
- $\alpha + \beta < 1$: long-run variance $\omega/(1 - \alpha - \beta)$, half-life $\ln 0.5 / \ln(\alpha + \beta)$
- $\alpha + \beta = 1$: IGARCH; EWMA is an IGARCH without ω

The likelihood of a GARCH model

- The returns are dependent, so the joint density is a product of conditional densities
  - $f(r_1, \ldots, r_T) = f(r_1) \prod_{t=2}^{T} f(r_t \mid \mathcal{F}_{t-1})$, with $r_t \mid \mathcal{F}_{t-1} \sim N(\mu, \sigma_t^2)$ for Normal innovations
- **Conditional log-likelihood**, $\theta = (\mu, \omega, \alpha, \beta)$:
  - $\ell(\theta) = -\frac{1}{2} \sum_{t=2}^{T} \left[\ln(2\pi) + \ln \sigma_t^2(\theta) + \frac{(r_t - \mu)^2}{\sigma_t^2(\theta)} \right]$
  - $\sigma_t^2(\theta)$ is computed recursively from a starting value σ_1^2 (for example the sample variance)
- **MLE** (maximum likelihood estimation): $\hat{\theta} = \arg\max_\theta \ell(\theta)$, under $\omega > 0$, $\alpha, \beta \geq 0$, $\alpha + \beta < 1$
  - no closed form: a numerical optimiser searches for the maximum; OLS is not possible, since σ_t^2 is not observed

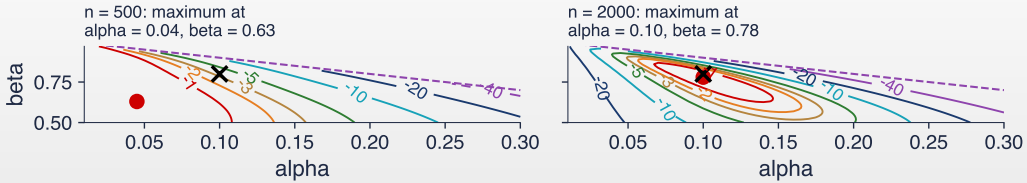
The likelihood of ARCH(1)



- Simulated ARCH(1) with $\omega = \alpha = 0.5$; $\ell(\alpha)$ with $\omega = 1 - \alpha$; each curve minus its maximum
- $n = 100$: $\hat{\alpha} = 0.432$ (standard error 0.127); $n = 1000$: $\hat{\alpha} = 0.491$ (0.035)
- Interpretation: more data make the curve sharper; its curvature at the maximum gives the standard error

TSA_ch5_simulation_likelihood

The likelihood of GARCH(1,1)



- Simulated GARCH(1,1) with $\omega = 0.1$, $\alpha = 0.1$, $\beta = 0.8$; ω set so that the unconditional variance equals the sample variance; contours: log-likelihood minus its maximum
- $n = 500$: maximum at $(0.045, 0.63)$, far from the truth; $n = 2000$: $(0.100, 0.78)$
- Interpretation: a long, flat ridge along which α and β trade off: GARCH needs long samples

TSA_ch5_simulation_likelihood

Estimation step by step and with the `arch` package

S&P 500, GARCH(1,1)-N	step by step (SE)	`arch`	classic SE	robust SE
μ	0.0614 (0.0098)	0.0615	0.0098	0.0100
ω	0.0253 (0.0030)	0.0254	0.0030	0.0048
α	0.1205 (0.0086)	0.1206	0.0086	0.0118
β	0.8600 (0.0091)	0.8599	0.0091	0.0125

- Step by step: $-\ell(\theta)$ written in Python and minimised with `scipy.optimize.minimize` under $\alpha + \beta < 1$
  - with the package: `arch_model(r, mean='Constant', vol='GARCH', p=1, q=1, dist='normal').fit()`
- SE (standard error), classic: from the inverse Hessian (the curvature of ℓ); robust: valid when z_t is not Normal Bollerslev and Wooldridge (1992)
- Interpretation: same estimates by both routes; the robust SE are about 1.4 times larger: with heavy-tailed z_t the classic SE overstate the precision

TSA_ch5_garch_estimation

Quasi-maximum likelihood and practical rules

- **QMLE** (quasi-maximum likelihood estimation): maximise the Normal likelihood even if z_t is not Normal
  - the estimates stay consistent if μ_t and σ_t^2 are correctly specified, but only the robust ("sandwich") SE are valid Bollerslev and Wooldridge (1992)
- Practical rules
  - returns in % (not decimals): the optimiser fails when ω is tiny; `arch` warns about badly scaled data
  - EUR/RON moves about 0.3% a day: we estimate on $10\,r_t$ and convert ω back (divide by 100)
  - use at least 1000–2000 daily observations; check the convergence flag of the optimiser
  - an estimate on the boundary ($\hat{\alpha} + \hat{\beta} = 1$ or $\hat{\alpha} = 0$) has no usual standard error: report it as such

Student-t and skewed-t innovations

- GARCH with Normal z_t explains only part of the kurtosis: S&P 500 returns 13.7, standardised residuals $\hat{z}_t = \hat{\varepsilon}_t/\hat{\sigma}_t$ still 4.7
  - the remaining heavy tails must come from the distribution of z_t
- **Standardised Student-t** with $\nu > 2$ degrees of freedom Bollerslev (1987)
  - $z_t = t_\nu \sqrt{(\nu-2)/\nu}$ has variance 1; kurtosis $3 + 6/(\nu-4)$ for $\nu > 4$; $\nu \to \infty$: the Normal distribution
  - ν is estimated together with $(\mu, \omega, \alpha, \beta)$: `dist='t'` in `arch`
- **Skewed t** Hansen (1994): one more parameter $\lambda \in (-1, 1)$; $\lambda < 0$: a longer left tail (`dist='skewt'`)
  - nested models: Normal $\subset$ t $\subset$ skewed t; compare them with LR (likelihood ratio) tests or AIC/BIC

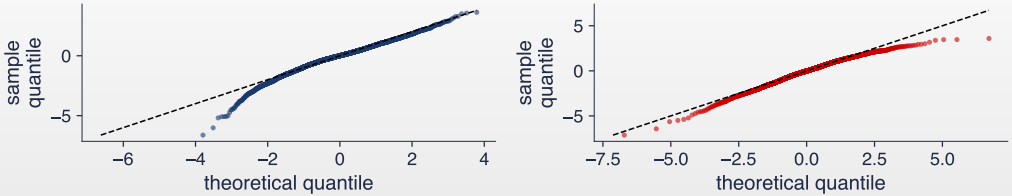
S&P 500: three innovation distributions

GARCH(1,1)	ω	α	β	ν	λ	log-lik.	$\alpha+\beta$
Normal	0.0254	0.121	0.860	–	–	−9218.0	0.980
Student-t	0.0157	0.123	0.872	6.26	–	−9062.5	0.995
skewed t	0.0153	0.120	0.872	6.92	−0.114	−9038.4	0.993

- LR statistic $2(\ell_1 - \ell_0)$, ℓ_1 and ℓ_0: the log-likelihoods of the larger and of the smaller model: t against Normal 311.0 (1 restriction, $\chi^2_{0.95}(1) = 3.84$); skewed t against t 48.1: both reject the smaller model
- $\hat{\nu} = 6.26$: tails far heavier than Normal; $\hat{\lambda} = -0.114$: large falls more frequent than large rises
- Interpretation: α and β hardly change; the innovation distribution matters for tail quantiles (VaR), less for σ_t

TSA_ch5_garch_estimation

QQ plots of the standardised residuals



- QQ (quantile–quantile) plot: sample quantiles of $\hat{z}_t$ against the quantiles of the assumed distribution; a straight line means a good fit
- Left, GARCH-N against the Normal: 0.1% quantile -4.81, against -3.09 in theory; right, GARCH-t against the standardised t ($\hat{\nu} = 6.3$): -4.79 against -4.19
- Interpretation: the t fits much better, but the largest falls are still deeper: the left tail is the problem (asymmetry, Section 7)

TSA_ch5_garch_estimation

Recap: Estimation by maximum likelihood

- $\ell(\theta)$ = sum of conditional log-densities; $\sigma_t^2(\theta)$ computed recursively; numerical maximisation
- The likelihood has a ridge in (α, β): long samples; report robust SE
- Student-t or skewed-t innovations for the remaining heavy tails

GARCH(1,1)-t estimates

	from	ω	α	β	ν	$\alpha+\beta$	half-life	long-run vol.	sample vol.
S&P 500	2000	0.0157	0.123	0.872	6.26	0.9947	131	27.3	19.2
BET	2000	0.0448	0.192	0.799	5.01	0.9907	74	34.7	22.1
EUR/RON	2005	0.0000	0.123	0.877	4.20	1.000	–	–	4.4
Bitcoin	2014	0.1937	0.109	0.891	3.18	1.000	–	–	66.6

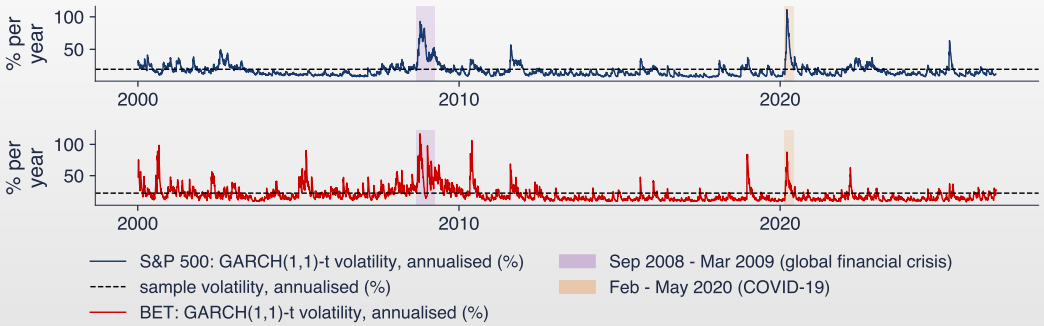
- Daily log returns in % to 18 September 2026; half-life in days; volatilities annualised with the actual number of observations per year (Bitcoin: 365), in %
- S&P 500 and BET: $\alpha+\beta$ 0.9947 and 0.9907, half-lives of 131 and 74 trading days; BET reacts more to news ($\alpha = 0.192$)
- EUR/RON and Bitcoin: $\hat{\alpha}+\hat{\beta}=1$ on the boundary: IGARCH; every $\hat{\nu}$ between 3.18 and 6.26: heavy tails everywhere

TSA_ch5_volatility_markets

Interpreting IGARCH and the long-run volatility

- EUR/RON: $\hat{\omega} \approx 0$ and $\hat{\beta} = 0.88$: an EWMA with $\lambda \approx 0.88$; Bitcoin: $\hat{\beta} = 0.89$
  - no half-life and no long-run volatility: the volatility level drifts like a random walk (Chapter 3)
  - EUR/RON: a managed rate whose volatility fell from one regime to the next (the BNR policy changes over time)
- **Question for the room**: S&P 500: long-run volatility 27.3%, sample volatility 19.2%. Is the model wrong?
  - **Answer**: not necessarily: $\bar{\sigma}^2 = \omega/(1 - \alpha - \beta)$ divides by $1 - \alpha - \beta = 0.0053$, a small and imprecise number
  - the Normal GARCH gives 18.1%: the long-run level is the least reliable output of a persistent GARCH

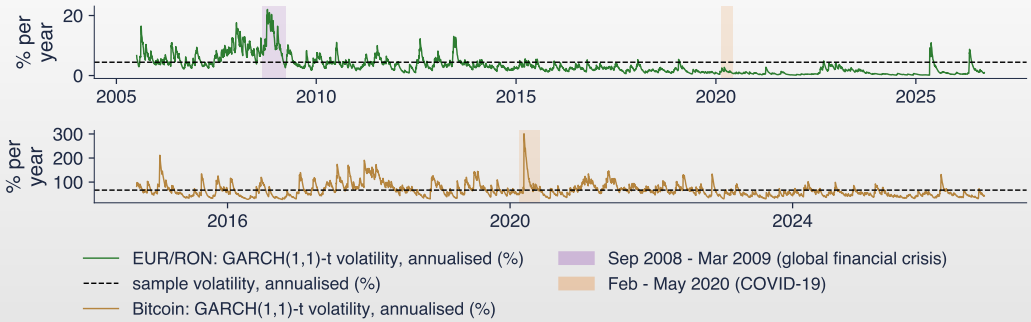
Conditional volatility: S&P 500 and BET



- GARCH(1,1)-t volatility $\hat{\sigma}_t$, annualised; dashed: the sample volatility
- S&P 500: peaks 93% (16 October 2008) and 111% (17 March 2020); BET: 117% (15 October 2008) and 88% (18 March 2020); medians 14.3% and 15.6%

TSA_ch5_volatility_markets

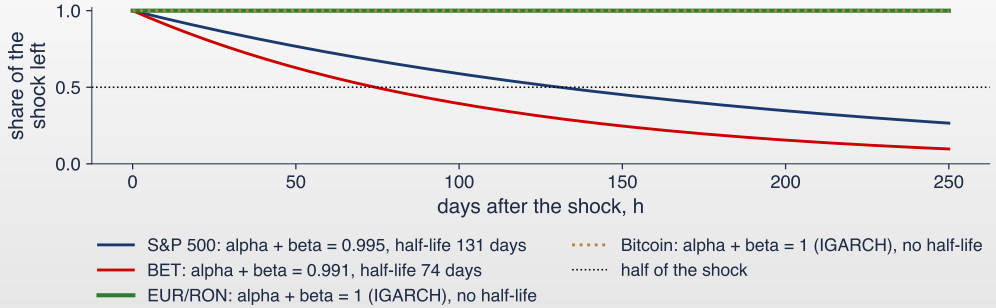
Conditional volatility: EUR/RON and Bitcoin



- EUR/RON: 22% at the peak of the 2008 crisis (13 October 2008), only 3% in 2020; on 18 September 2026: 1.0%
- Bitcoin: 301% on 13 March 2020; median 58.7%, 4.1 times the median of the S&P 500
- Interpretation: the two IGARCH series have volatility levels that wander: no single "normal" level

TSA_ch5_volatility_markets

How long does a volatility shock last?



- Share of a variance shock left after h days: $(\alpha + \beta)^h$; the half-life is where the curve crosses 1/2
- Interpretation: after a crisis the S&P 500 needs about half a year to halve the excess variance, the BET about three months; EUR/RON and Bitcoin never

TSA_ch5_volatility_markets

ARMA-GARCH: one model for the mean and the variance

- **ARMA(p,q)-GARCH(1,1)**: $r_t = c + \sum_{i=1}^{p} \phi_i r_{t-i} + \varepsilon_t + \sum_{j=1}^{q} \theta_j \varepsilon_{t-j}, \quad \varepsilon_t = \sigma_t z_t, \quad \sigma_t^2 = \omega + \alpha \varepsilon_{t-1}^2 + \beta \sigma_{t-1}^2$
  - the ARMA part is the model of Chapter 2; its errors are no longer i.i.d., they follow a GARCH
- Estimation: both parts at once, by maximum likelihood
  - in `arch`: `arch_model(r, mean='AR', lags=1, vol='GARCH', dist='t')` (AR means only; for MA terms use two steps or another package)
- Why not OLS for the mean and then GARCH?
  - the OLS coefficients are still consistent, but their usual SE assume a constant variance: the t tests of Chapter 2 are wrong
  - the joint maximum likelihood estimate gives less weight to stormy days, hence more precise $\hat{\phi}$ and valid intervals

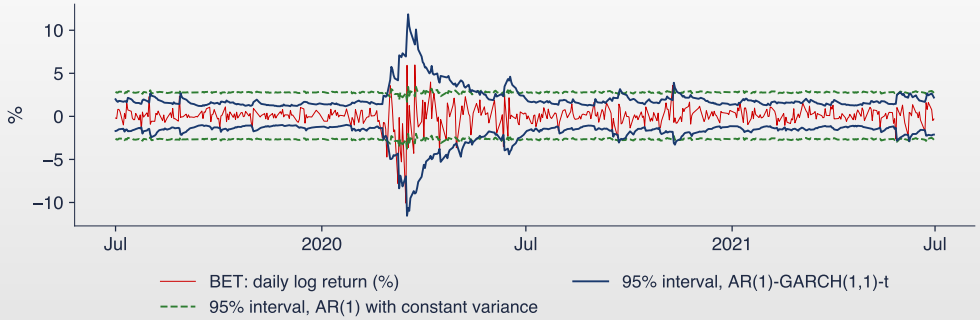
AR(1)-GARCH(1,1)-t on four series

	p (BIC)	$\hat{\phi}$, OLS (SE)	$\hat{\phi}$, AR-GARCH (SE)	$Q(10)$ of $\hat{z}_t$, const. mean	$Q(10)$ of $\hat{z}_t$, AR(1)	ΔBIC
S&P 500	1	−0.099 (0.012)	−0.054 (0.012)	21.9 (0.016)	15.1 (0.087)	−19.5
BET	1	0.109 (0.012)	0.091 (0.013)	114.6 (< 0.001)	28.6 (< 0.001)	−46.9
EUR/RON	3	0.170 (0.013)	0.031 (0.016)	10.5 (0.402)	8.7 (0.469)	4.2
Bitcoin	0	−0.022 (0.015)	−0.043 (0.014)	29.9 (< 0.001)	48.4 (< 0.001)	−9.5

- p: AR order chosen by BIC with a constant variance; SE of the AR-GARCH: robust; ΔBIC: AR(1)-GARCH-t minus GARCH-t with a constant mean (negative: AR(1) preferred)
- S&P 500: $\hat{\phi}$ shrinks from -0.099 to -0.054: the OLS value is driven by the stormy days of 2008 and 2020
- BET: a positive autocorrelation survives ($\hat{\phi} = 0.091$, $t = 6.8$): slow price adjustment in a less liquid market; EUR/RON: from 0.170 to 0.031, no longer significant
- Interpretation: the mean model and the variance model must be chosen together; the conclusions about ϕ can change

TSA_ch5_arma_garch

Forecast intervals that breathe



- BET, one-day-ahead 95% intervals: AR(1) with constant variance (width 5.4 points every day) and AR(1)-GARCH(1,1)-t (width from 2.2 to 22.5 points in this window)

TSA_ch5_arma_garch

Interpreting the two intervals

- Over the whole sample both miss about 5% of the days: 5.0% and 5.2%
  - the constant interval is right on average, but wrong almost every day
- In 2017 (calm): the constant interval misses 0.4% of the days, the GARCH interval 2.0%
  - in March–April 2020: 33.3% against 9.5%: the misses of the constant interval cluster in crises
- Chapter 2 gave intervals whose width depends only on the horizon; with GARCH it also depends on the current state

Recap: Four markets and ARMA-GARCH

- $\alpha + \beta$ close to 1 everywhere; EUR/RON and Bitcoin are IGARCH
- The long-run volatility is imprecise when $\alpha + \beta$ is near 1
- ARMA-GARCH: the mean of Chapter 2 with GARCH errors; joint MLE; intervals that follow the volatility

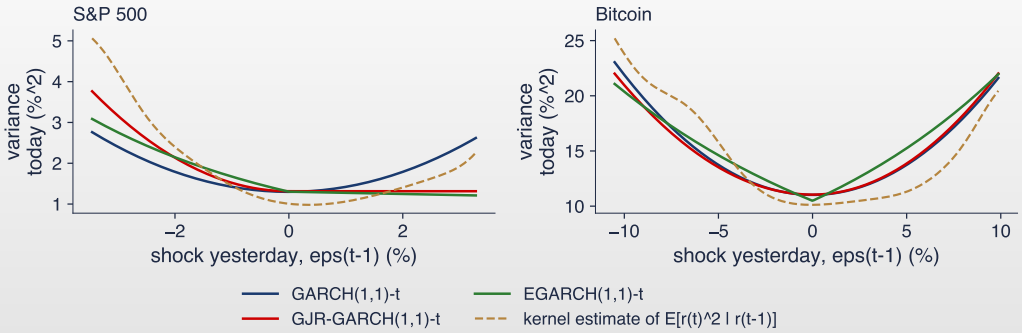
The leverage effect and the news impact curve

- **Leverage effect**: falls in stock prices raise volatility more than rises of the same size Christie (1982)
  - a lower share price raises the debt-to-equity ratio, so the equity becomes riskier; news of higher risk also lowers prices at once
- GARCH(1,1) cannot see it: σ_t^2 depends on ε_{t-1}^2, not on the sign of ε_{t-1}
  - the QQ plot and the skewed t ($\hat{\lambda} < 0$) already pointed to the left tail
- **NIC** (news impact curve) Engle and Ng (1993): σ_t^2 as a function of ε_{t-1}, with σ_{t-1}^2 fixed at the unconditional variance; for GARCH, a symmetric parabola

GJR-GARCH and EGARCH

- **GJR-GARCH** Glosten, Jagannathan and Runkle (1993): $\sigma_t^2 = \omega + (\alpha + \gamma\, I_{t-1})\, \varepsilon_{t-1}^2 + \beta\sigma_{t-1}^2$, with $I_{t-1} = 1$ if $\varepsilon_{t-1} < 0$, else 0
  - good news: slope α; bad news: $\alpha + \gamma$; leverage effect: $\gamma > 0$; in `arch`: `o=1`
  - persistence $\alpha + \beta + \gamma/2$ for symmetric z_t (half of the shocks are negative)
- **EGARCH** (exponential GARCH) Nelson (1991): $\ln \sigma_t^2 = \omega + \alpha\big(|z_{t-1}| - E|z_{t-1}|\big) + \gamma z_{t-1} + \beta \ln \sigma_{t-1}^2$
  - z_{t-1}: yesterday's standardised shock; $E|z_{t-1}|$: its mean absolute value ($\sqrt{2/\pi}$ for the Normal distribution); α: size effect; γ: sign effect
  - the logarithm keeps $\sigma_t^2 > 0$ without sign restrictions; leverage effect: $\gamma < 0$; persistence: β
- Both have one parameter more than GARCH(1,1); test $\gamma = 0$ with its t statistic or with an LR test, $\chi^2(1)$

News impact curves: S&P 500 and Bitcoin



- Model curves: σ_t^2 against ε_{t-1}, with σ_{t-1}^2 at the sample variance; dashed: a kernel (Nadaraya–Watson) estimate of $E[r_t^2 \mid r_{t-1}]$, without a model
- S&P 500, GJR: after a fall of 2%, tomorrow's variance is 1.62 times the variance after a rise of 2%; Bitcoin: ratio 1.00, a symmetric curve

TSA_ch5_asymmetry

Asymmetry in four markets

	GJR α	GJR γ	$t(\gamma)$	LR	p	ΔBIC	EGARCH γ	$t(\gamma)$
S&P 500	0.000	0.205	10.0	235.7	< 0.001	-226.8	-0.165	-15.2
BET	0.172	0.043	2.1	4.9	0.026	3.9	-0.028	-2.6
EUR/RON	0.103	0.039	3.0	9.3	0.002	-0.7	-0.040	-2.5
Bitcoin	0.113	-0.013	-0.7	0.7	0.416	7.7	0.015	1.2

- Student-t innovations; robust t statistics; LR $= 2(\ell_{GJR} - \ell_{GARCH}) \sim \chi^2(1)$; ΔBIC: GJR minus GARCH (negative: GJR preferred)
- S&P 500: GJR $\hat{\alpha} = 0.000$: only bad news raises volatility; a very strong leverage effect
- BET: weak, significant by LR, but BIC prefers GARCH; Bitcoin: none; EUR/RON: $r_t > 0$ means a weaker leu, so the "bad news" of GJR are days when the leu strengthens
- Interpretation: leverage is a property of mature equity markets, not of every financial series

TSA_ch5_asymmetry

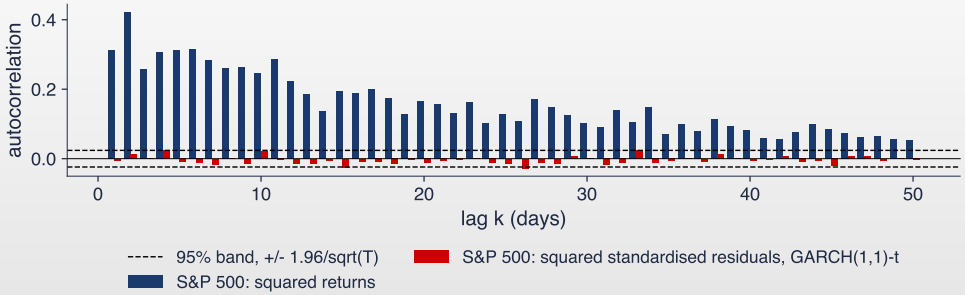
Recap: Asymmetry

- GJR: an extra slope γ for negative shocks; EGARCH: a sign term γz_{t-1} in $\ln \sigma_t^2$
- The news impact curve shows the asymmetry at a glance
- Strong in the S&P 500, weak in the BET, absent in Bitcoin

Checking a fitted model

- If the model is right, the **standardised residuals** $\hat{z}_t = (r_t - \hat{\mu}_t)/\hat{\sigma}_t$ are close to i.i.d. with mean 0 and variance 1
  - Ljung–Box $Q(10)$ on $\hat{z}_t$: no autocorrelation left in the mean (the check of Chapter 2)
  - Ljung–Box $Q(10)$ on $\hat{z}_t^2$ and ARCH-LM on $\hat{z}_t$: no ARCH effects left
  - QQ plot of $\hat{z}_t$: is the assumed distribution right?
- **Sign-bias test** Engle and Ng (1993): regress $\hat{z}_t^2$ on I_{t-1}, $I_{t-1}\hat{\varepsilon}_{t-1}$ and $(1 - I_{t-1})\hat{\varepsilon}_{t-1}$
  - joint test $TR^2 \sim \chi^2(3)$: a rejection means that the sign of past shocks still predicts the variance
- A passed diagnostic does not prove the model right; a failed one shows where it is wrong

What GARCH removes



- S&P 500: ACF of r_t^2: 0.31 at lag 1, still 0.053 at lag 50; ACF of $\hat{z}_t^2$ (GARCH(1,1)-t): -0.005 at lag 1, at most 0.028 in absolute value; band ± 0.024
- Interpretation: four parameters absorb almost all the volatility clustering of 27 years of daily data

TSA_ch5_diagnostics

Diagnostics on real data

S&P 500	$Q(10)$ (p)	$Q(10)$ of squares (p)	ARCH-LM(5) (p)
returns r_t	96.3 (<0.001)	6133 (<0.001)	1684 (<0.001)
$\hat{z}_t$, GARCH-N	21.2 (0.020)	16.4 (0.089)	7.0 (0.222)
$\hat{z}_t$, GARCH-t	21.9 (0.016)	13.6 (0.192)	4.8 (0.438)

- GARCH removes the ARCH effects ($Q(10)$ of squares: from 6133 to 13.6); a little autocorrelation remains in the mean (constant mean; an AR(1) helps, Section 6)
- Sign bias for the GARCH-t of the S&P 500: $\chi^2(3) = 40.7$ (p < 0.001): the asymmetry of Section 7 is missing
- $Q(10)$ of $\hat{z}_t^2$, GARCH-t: BET 8.3 (p $= 0.596$), Bitcoin 4.7 (p $= 0.910$), EUR/RON 0.004: suspiciously small (next slide)

TSA_ch5_diagnostics

Case study: the EUR/RON on 6 May 2025

- In April 2025 the reference rate moved by 0.0038% a day on average: the BNR held the leu almost fixed
  - the IGARCH volatility fell to $\hat{\sigma}_t = 0.0078\%$; then the rate rose by 1.20% (two days after the first round of the presidential election) and by 1.21% the next day
- Standardised residual: $\hat{z}_t = 154$, a "154-sigma" day; the next largest is 13.2
  - $Q(10)$ of $\hat{z}_t^2$: 0.004 with this day, 19.2 ($p = 0.038$) without it: one value hides the remaining ARCH effects
- Lesson: a managed rate is not a free market; GARCH cannot foresee a policy decision; always plot $\hat{z}_t$ before trusting a test



The National Bank of Romania, Bucharest

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TSA_ch5_diagnostics

Nine models for the S&P 500

- AIC Akaike (1974) and BIC Schwarz (1978) as defined on ARCH(q) and its limits; compare models fitted on the same data

	Normal		Student-t		skewed t	
	log-lik.	ΔBIC	log-lik.	ΔBIC	log-lik.	ΔBIC
GARCH(1,1)	−9218.0	648.9	−9062.5	346.7	−9038.4	307.4
GJR-GARCH(1,1)	−9086.5	394.7	−8944.6	119.9	−8905.1	49.6
EGARCH(1,1)	−9075.6	372.9	−8922.1	74.8	−8880.3	0.0

- ΔBIC: the difference from the best model, EGARCH-skewed t
- From the Normal GARCH: asymmetry alone (Normal EGARCH) lowers BIC by 276, heavy tails alone (GARCH-skewed t) by 341, both by 649
- Interpretation: with 6 714 observations even small gains are "significant"; the out-of-sample comparison of Section 9 is the real test

TSA_ch5_diagnostics

Recap: Diagnostics and model choice

- Standardised residuals: no autocorrelation in $\hat{z}_t$ and $\hat{z}_t^2$, the right distribution, no sign bias
- One extreme $\hat{z}_t$ can hide everything: plot before testing
- AIC/BIC on the same data; in-sample fit is not forecasting ability

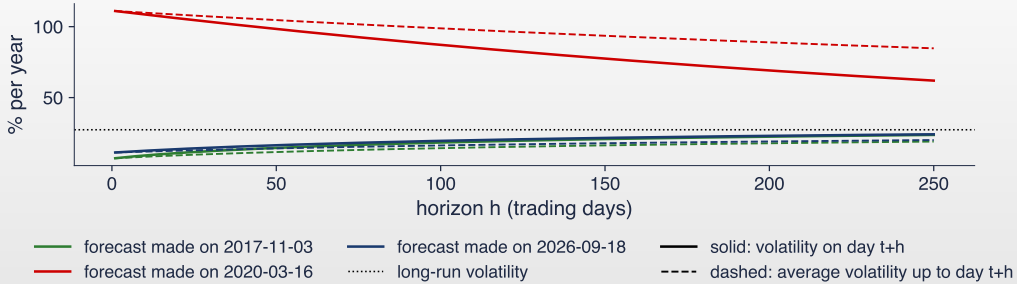
Multi-step forecasts

- One step: $\sigma^2_{t+1} = \omega + \alpha\varepsilon^2_t + \beta\sigma^2_t$ is known at the end of day t
  - two steps: $E_t[\sigma^2_{t+2}] = \omega + \alpha E_t[\varepsilon^2_{t+1}] + \beta\sigma^2_{t+1} = \omega + (\alpha+\beta)\sigma^2_{t+1}$, since $E_t[\varepsilon^2_{t+1}] = \sigma^2_{t+1}$
- By induction: $E_t[\sigma^2_{t+h}] = \bar{\sigma}^2 + (\alpha+\beta)^{h-1}(\sigma^2_{t+1} - \bar{\sigma}^2)$
  - the forecast converges to the long-run variance, like an AR(1) forecast converges to its mean (Chapter 2)
- Variance of the H-day return: the sum of the daily forecasts (the returns are uncorrelated)
  - $\text{Var}_t(r_{t+1} + \cdots + r_{t+H}) = \sum_{h=1}^{H} E_t[\sigma^2_{t+h}]$, not $H\sigma^2_{t+1}$
  - the square-root-of-time rule $\sigma_{t+1}\sqrt{H}$ is too high in a crisis and too low in a calm period

Worked example: the S&P 500 on 18 September 2026

- GARCH(1,1)-t: $\alpha + \beta = 0.9947$, $\bar{\sigma}^2 = 2.972$; forecast for the next day $\sigma^2_{t+1} = 0.509$ (below the long-run level)
- Step by step
  - day 10: $2.972 + 0.9947^9(0.509 - 2.972) = 2.972 + 0.953 \times (0.509 - 2.972) = 0.624$
  - 10-day variance: the sum of the ten daily forecasts $= 5.67$, so the 10-day volatility is 2.38%
  - square-root-of-time rule: $\sqrt{10 \times 0.509} = 2.26\%$: too low, because volatility is expected to rise towards its long-run level

The term structure of volatility



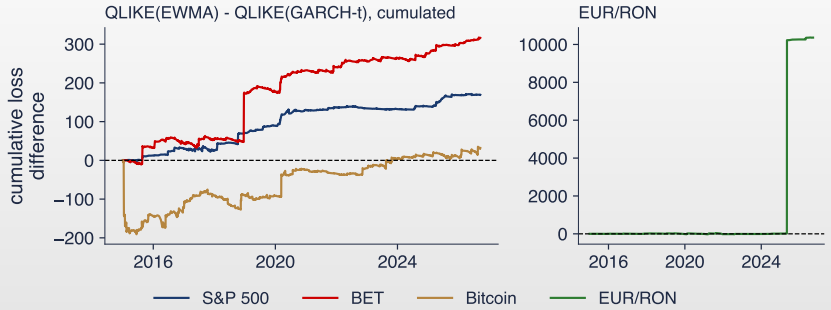
- S&P 500, GARCH(1,1)-t on all data; forecasts made on a calm day (3 November 2017), at the COVID-19 peak (16 March 2020) and on 18 September 2026
- Calm day: 7.3% tomorrow, 19.0% on average over a year; COVID-19 peak: 111.0% tomorrow, still 84.7% on average over a year
- Interpretation: all curves move towards the long-run level (27.3%) at the same slow speed; the slope tells whether we are in a calm period or in a storm

TSA_ch5_forecasts

How to evaluate a volatility forecast

- The true σ_t^2 is never observed: we compare the forecast h_t with a noisy **proxy**, here r_t^2
  - $E_{t-1}[r_t^2] = \sigma_t^2$ (with $\mu \approx 0$), but r_t^2 is very noisy: a low R^2 even for a perfect forecast Andersen and Bollerslev (1998)
- **QLIKE** loss Patton (2011): $L(r_t^2, h_t) = \dfrac{r_t^2}{h_t} + \ln h_t$ (up to a constant); the smaller the mean, the better
  - it ranks forecasts correctly even with a noisy proxy; it is minus the Normal log-likelihood; it punishes forecasts that are too low
- **Out of sample** (Chapter 0): from 2015, each forecast uses only data up to the day before; the models are re-estimated every 250 days
- **DM test** Diebold and Mariano (1995): H_0: $E[d_t] = 0$, with $d_t = L_t^A - L_t^B$ the loss difference of two models A and B
  - $\mathrm{DM} = \bar{d}/\widehat{\mathrm{SE}}(\bar{d})$, approximately $N(0,1)$ under H_0; $\bar{d}$: the mean of d_t; $\mathrm{DM} < -1.96$: A is better
  - $\widehat{\mathrm{SE}}$: HAC (heteroskedasticity and autocorrelation consistent) standard error

GARCH against EWMA, out of sample



- One-day-ahead variance forecasts from 2015 to 18 September 2026: GARCH(1,1)-t (re-estimated every 250 days) against EWMA with $\lambda = 0.94$; a rising curve means that GARCH-t has the smaller QLIKE

TSA_ch5_forecasts

Interpreting the forecast comparison

2015–2026	days	QLIKE GARCH-t	QLIKE GJR-t	QLIKE EWMA	DM t, GARCH-t against EWMA (p)
S&P 500	2 944	0.761	0.733	0.819	-3.69 (<0.001)
BET	2 926	0.680	0.674	0.788	-2.22 (0.027)
EUR/RON	2 939	2.456	3.276	5.982	-1.02 (0.310)
Bitcoin	4 279	3.383	3.401	3.390	-0.20 (0.845)

- S&P 500 and BET: GARCH-t beats EWMA ($t < -1.96$); GJR-t has the smallest QLIKE (the leverage effect pays off out of sample)
- Bitcoin: no difference ($t = -0.20$): its GARCH is an IGARCH, which is almost an EWMA
- EUR/RON: a much smaller mean QLIKE for GARCH, yet $t = -1.02$: 98% of the total difference comes from one day, 6 May 2025, when EWMA forecast a near-zero variance
- Interpretation: QLIKE punishes forecasts that are too low; the DM test protects us from conclusions built on a single day

TSA_ch5_forecasts

Case study: Hansen and Lunde (2005)

- **Question**: does anything beat a GARCH(1,1)? Hansen and Lunde (2005)
  - 330 ARCH-type models (asymmetric, long memory, different innovations), compared out of sample on the Deutsche Mark/US dollar exchange rate and on the returns of one US stock
  - the proxy: realised variance from intraday returns, more precise than r_t^2
- **Finding**: for the exchange rate, no model beat GARCH(1,1) significantly; for the stock, models with a leverage effect did better
  - the same pattern as our table: asymmetry matters for equities, not for currencies or Bitcoin
- **Legacy**: GARCH(1,1) is the benchmark every new volatility model, including machine learning (Chapter 9), must beat out of sample

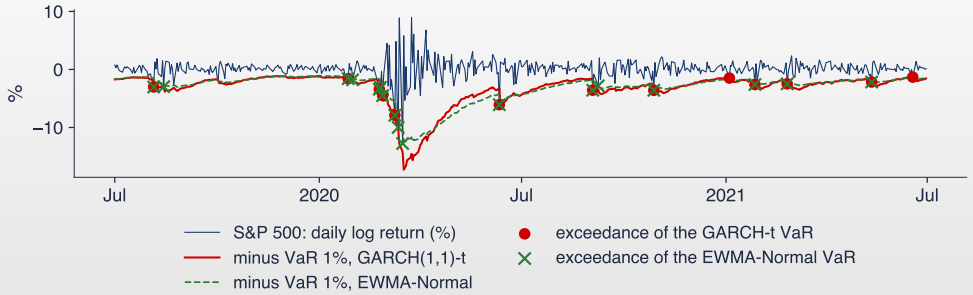
Recap: Variance forecasts

- $E_t[\sigma^2_{t+h}] = \bar{\sigma}^2 + (\alpha+\beta)^{h-1}(\sigma^2_{t+1} - \bar{\sigma}^2)$; H-day variance = sum of the daily forecasts
- Compare forecasts out of sample with QLIKE and the Diebold–Mariano test
- GARCH beats EWMA where volatility mean-reverts

Conditional VaR

- **VaR** (value at risk) at level α: the loss exceeded with probability α, $\text{VaR}_\alpha = -q_\alpha(r_{t+1})$, with q_α the α-quantile
  - here $\alpha = 1\%$: VaR 1%, a loss exceeded on one day in a hundred
- With $r_{t+1} = \mu + \sigma_{t+1} z_{t+1}$: $\text{VaR}_{t+1} = -(\mu + \sigma_{t+1}\, q_\alpha(z))$
  - $q_\alpha(z)$: the α-quantile of the innovation z; Normal: $q_{0.01} = -2.326$
  - standardised t: $q_{0.01} = t_\nu^{-1}(0.01)\sqrt{(\nu-2)/\nu}$, t_ν^{-1}: the quantile function of the Student distribution with ν degrees of freedom
  - the VaR moves every day with σ_{t+1}: high in storms, low in calm periods
- **Worked example**: S&P 500 on 18 September 2026, GARCH(1,1)-t: $\hat{\mu} = 0.079$, $\sigma_{t+1} = 0.714$, $\hat{\nu} = 6.26$
  - $t_{\hat{\nu}}^{-1}(0.01) = -3.100$, $\sqrt{(\hat{\nu}-2)/\hat{\nu}} = 0.825$, so $q_{0.01} = -3.100 \times 0.825 = -2.557$
  - VaR 1% $= -(0.079 + 0.714 \times (-2.557)) = 1.75\%$
  - with Normal z_t: 1.58%; over 10 days (sum of the variances, μ ignored): $2.557 \times \sqrt{5.67} = 6.09\%$

VaR 1% through the COVID-19 crash



- S&P 500, July 2019 – June 2021 (505 days): one-day VaR 1% from GARCH(1,1)-t (re-estimated every 250 days) and from EWMA with Normal z_t
- Exceedances (return below −VaR): GARCH-t 13, EWMA-Normal 17, about 5 expected; both models react, but only after the first large falls of March 2020

TSA_ch5_forecasts

Exceedances 2015–2026

VaR 1%	days	expected	GARCH-t	EWMA-Normal
S&P 500	2 944	29	49 (1.66%)	65 (2.21%)
BET	2 926	29	34 (1.16%)	67 (2.29%)
EUR/RON	2 939	29	35 (1.19%)	63 (2.14%)
Bitcoin	4 279	43	64 (1.50%)	93 (2.17%)

- A correct VaR 1% is exceeded on 1% of the days, and the exceedances do not cluster
- EWMA-Normal: 2.1–2.3% everywhere: the Normal quantile is too small for heavy tails
- GARCH-t: 1.2–1.7%: much closer to 1%; the highest for the S&P 500, where the leverage effect is missing
- Are the differences significant? The test of Kupiec (1995) compares the number of exceedances with a binomial distribution

TSA_ch5_forecasts

Recap: VaR 1% from a GARCH model

- $\text{VaR}_{t+1} = -(\mu + \sigma_{t+1} q_\alpha(z))$: a dynamic volatility and the right quantile
- Normal quantiles underestimate the risk; Student-t is much closer to 1%
- Several series at once (covariances of returns): multivariate GARCH, Chapter 14 (self-study)

Possible contribution of AI

- **Code**: a first draft of a script that runs ARCH-LM, fits GARCH, GJR and EGARCH with `arch` on many series and tabulates the results
- **Explanation**: a second explanation of the ARMA(1,1) form of GARCH, or of an `arch` output table
- **Exploration**: volatility of all BVB stocks, or of the leu against several currencies
- Example prompt
  - `Write Python code that loads daily BET closes, computes log returns in percent, runs the ARCH-LM test with 5 lags, fits AR(1)-GARCH(1,1) with Student-t innovations using the arch package, and plots the annualised conditional volatility.`

Checks you must run

- Units: returns in %, not decimals; rescale a series with a very small variance (EUR/RON) and convert ω back
- The persistence: $\alpha + \beta$ for GARCH, $\alpha + \beta + \gamma/2$ for GJR, β for EGARCH
- Boundary estimates ($\alpha + \beta = 1$): no half-life and no long-run volatility; do not report infinite numbers
- Calendars: Bitcoin trades 7 days a week, the BET five; annualise each series with its own frequency
- Out-of-sample forecasts use only past data; the VaR level is written as VaR 1%
- Every cited reference: it must exist; check the DOI

Key takeaways

- Returns: an almost unpredictable mean and a predictable variance, $r_t = \mu_t + \sigma_t z_t$; test with the ACF of r_t^2 and ARCH-LM
- GARCH(1,1) is an ARMA(1,1) for ε_t^2; $\alpha + \beta$ close to 1; IGARCH and EWMA when $\alpha + \beta = 1$
- Estimate by maximum likelihood with `arch`, report robust SE; Student-t innovations for the tails
- ARMA-GARCH: the mean of Chapter 2 and the variance of this chapter estimated together
- Equity indices need asymmetry (GJR, EGARCH); check $\hat{z}_t$ and $\hat{z}_t^2$ after every fit
- Forecasts revert to the long-run level; compare them out of sample with QLIKE and DM; VaR 1% $= -(\mu + \sigma_{t+1} q_{0.01}(z))$

Key formulas

Quantity	Formula
ARCH-LM	$\hat{\varepsilon}_t^2 = b_0 + \sum_{i=1}^{q} b_i \hat{\varepsilon}_{t-i}^2 + u_t, \quad \text{LM} = nR^2 \sim \chi^2(q)$
ARCH(q)	$\sigma_t^2 = \omega + \sum_{i=1}^{q} \alpha_i \varepsilon_{t-i}^2, \quad \bar{\sigma}^2 = \omega/(1 - \sum \alpha_i)$
GARCH(1,1)	$\sigma_t^2 = \omega + \alpha\varepsilon_{t-1}^2 + \beta\sigma_{t-1}^2, \quad \bar{\sigma}^2 = \omega/(1-\alpha-\beta), \quad h_{1/2} = \ln 0.5 / \ln(\alpha+\beta)$
EWMA	$\sigma_t^2 = \lambda\sigma_{t-1}^2 + (1-\lambda) r_{t-1}^2$
GJR, EGARCH	$\sigma_t^2 = \omega + (\alpha + \gamma I_{t-1})\varepsilon_{t-1}^2 + \beta\sigma_{t-1}^2; \quad \ln\sigma_t^2 = \omega + \alpha(\lvert z_{t-1}\rvert - E\lvert z_{t-1}\rvert) + \gamma z_{t-1} + \beta \ln\sigma_{t-1}^2$
Log-likelihood	$\ell = -\frac{1}{2}\sum_t [\ln 2\pi + \ln\sigma_t^2 + (r_t - \mu_t)^2/\sigma_t^2]$
Forecast	$E_t[\sigma_{t+h}^2] = \bar{\sigma}^2 + (\alpha+\beta)^{h-1}(\sigma_{t+1}^2 - \bar{\sigma}^2)$
QLIKE, VaR 1%	$L = r_t^2/h_t + \ln h_t; \quad \text{VaR}_{t+1} = -(\mu + \sigma_{t+1}\, q_{0.01}(z))$

Self-assessment

- **Question**: $\omega = 0.05$, $\alpha = 0.05$, $\beta = 0.90$. What are the long-run variance and the half-life?
  - **Answer**: $0.05/0.05 = 1$; $\ln 0.5 / \ln 0.95 = 13.5$ days
- **Question**: an ARCH-LM regression with $q = 5$ on 1000 residuals gives $R^2 = 0.004$. Are there ARCH effects at 5%?
  - **Answer**: LM $= 1000 \times 0.004 = 4 < 11.07$: H_0 is not rejected, no evidence of ARCH effects at 5% (which does not prove they are absent)
- **Question**: GJR gives $\hat{\alpha} = 0$ and $\hat{\gamma} = 0.2$. What happens to the variance after a rise of 3%?
  - **Answer**: nothing beyond the decay $\beta\sigma_t^2$: only negative shocks raise the variance
- Next: Chapter 6, VAR models and Granger causality (several series at once)

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